

BACKEND WORKFLOW

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Introduction

Backend development refers to the server-side functionality of an application. It is responsible for handling business logic, managing databases, processing requests, implementing authentication, and ensuring communication between the frontend and the database.

The backend plays a critical role in maintaining application performance, security, and data management.

Backend Workflow Overview

The backend workflow represents the sequence of operations that occur when a user interacts with an application.

Basic Workflow Diagram

User → Frontend → Backend Server → Database



Response

Step-by-Step Backend Workflow

1. Client Request

The workflow begins when a user performs an action through the frontend application, such as:

- Logging in
- Registering
- Fetching data
- Submitting forms

The frontend sends an HTTP request to the backend server.

Example

POST /login

2. API Endpoint Handling

The backend server receives the request through predefined API routes or endpoints.

Example

```
app.post('/login', loginUser);
```

The request is then forwarded to the appropriate controller or handler function.

3. Request Processing

Controllers process incoming requests by:

- Validating input data
- Executing application logic
- Calling required services
- Preparing responses

Example

```
function loginUser(req, res) {  
  // authentication logic  
}
```

4. Business Logic Execution

The business logic layer performs operations such as:

- User verification
- Data processing
- Calculations
- Rule validation
- Authorization checks

This layer ensures the application behaves according to functional requirements.

5. Database Communication

The backend interacts with databases to:

- Store information
- Retrieve records
- Update existing data
- Delete records

Commonly used databases include:

- MongoDB
- MySQL
- PostgreSQL

REST API Communication

REST APIs enable communication between the frontend and backend using HTTP methods.

Authentication Flow

Authentication verifies the identity of users before granting access to protected resources.

Authentication Process

User Login Request



Backend Validates Credentials



Token/Session Generated



Access Granted

Common Authentication Methods

- JWT (JSON Web Token)
- Session-Based Authentication
- OAuth Authentication

Backend Architecture

A standard backend architecture is structured into multiple layers for better scalability and maintainability.

Architecture Flow

Routes → Controllers → Services → Database

Components

Routes

Define API endpoints and direct requests.

Controllers

Handle request and response processing.

Services

Contain business logic and core operations.

Database Layer

Manages data storage and retrieval.

Conclusion

Backend workflow is essential for handling application functionality, managing data, processing requests, implementing authentication, and ensuring secure communication between the frontend and database systems.

A well-structured backend architecture improves:

- Scalability
- Security
- Performance
- Maintainability
- Reliability

Backend development forms the foundation of modern web and software applications.